UDC TBD
DOI: To be assigned by the publisher

HIERARCHICAL MAINTENANCE IN TAIL ASSIGNMENT: FORMULATION AND OPTIMIZATION

Draft placeholder: Write the English abstract after the results are validated. Target 200–250 words and 1,800–2,500 characters including spaces. State the maintenance generalization, formulation improvements, matched comparison protocol, and supported findings.

K e y w o r d s: ***mixed-integer programming, aircraft routing, cumulative exposure, calendar constraints, valid inequalities, computational evaluation***

AMS 2020 classification: 90C11, 90B35

Draft placeholder: This is a journal-specific skeleton, not a submission-ready manuscript. Author metadata, UDC, abstracts, references, and results need verification. Target main-text length: 5,000–8,000 words, preferably 6,800.

1. Introduction

Draft placeholder: Target: 650 words. Establish the maintenance-generalization problem, practical motivation, research gap, aim, and three research questions. Introduce the compact formulation of (Khaled, Minoux, Mousseau, Michel, & Ceugniet, 2018) as a baseline, not as the current state of the art. Keep earlier correctness work to one brief contextual paragraph without counterexamples.

2. Related work and comparison framework

2.1 Compact, network, and route-based formulations

Aircraft planning problems differ in both the decisions optimized and the maintenance obligations represented. Fleet assignment selects equipment types, whereas tail assignment associates scheduled flights with individual aircraft. Maintenance routing couples those assignments to opportunities for servicing the aircraft. Check scheduling additionally allocates maintenance periods and resources, and task allocation determines the work performed within each check. Consequently, a method that schedules maintenance for existing rotations is not automatically a competing solution method for unrestricted tail assignment. These distinctions determine which published results can support a direct comparison and which instead provide methodological context.

For the present study, (Khaled et al., 2018) provides the historical compact tail-assignment reference. The article's relevance is its direct connection to the assignment-based representation extended here; its publication date does not make it a sufficient description of contemporary maintenance optimization. Three formulation paradigms organize the broader comparison. Compact assignment models place the principal decisions on flight–aircraft pairs and use additional conditions for routing and maintenance. Network models represent feasible aircraft transitions explicitly. Route-based models select complete flight sequences, often requiring column generation when the route set cannot be enumerated. Their computational characteristics depend on preprocessing, relaxation strength, and the solution procedure, not only on the number of variables in the initial model.

Accordingly, the comparison should not rank these paradigms from smallest to largest or assume that compactness establishes efficiency. A network model may require more variables while expressing continuity more directly. A route-based method can exploit route feasibility during pricing, but its implementation and performance depend on the pricing and branching mechanisms. For this article, an alternative formulation is a direct comparator only after its objective, coverage requirements, aircraft states, and maintenance semantics have been aligned with the compact formulation.

2.2 Maintenance generalizations and the research gap

Joint maintenance scheduling and tail assignment already have substantial precedents. (Torres Sanchez, Boyacı, & Zografos, 2020) develop a framework combining multi-objective mixed-integer formulations with an iterative solution method. Their author manuscript models multiple maintenance types, aircraft-specific airworthiness states, maintenance duration, and workshop resource demands. Maintenance opportunities are represented through events and intervals derived from the flight schedule. The scheduling formulation accounts for initial airworthiness and resource availability, while the tail-assignment extension allows changes to the supplied allocation. Its lexicographic objectives include maintenance violations and resource-related criteria. This work rules out treating either multiple maintenance classes or the integration of maintenance scheduling with tail assignment as a new contribution by itself.

It also illustrates why numerical comparisons require care. A formulation that explicitly minimizes violations and modifies an existing tail allocation does not share the same objective and feasible region as a hard-feasibility assignment-cost model. Workshop resource vectors are not interchangeable with a single station–day slot capacity. These features should be described as differences in scope, not reduced to an assertion that one model is universally more realistic. The maintenance regime, temporal resolution, and resource interpretation must be stated before comparing solution quality or scale.

Recent developments extend the problem in other directions. (Glomb, Liers, & Rösel, 2023) integrate predictive-maintenance information into tail assignment through failure scenarios and a stochastic mixed-integer formulation. Their published abstract describes a Benders-based solution procedure and comparisons with preventive, reactive, and passive

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strategies. This contribution concerns uncertainty and recovery costs, rather than only deterministic exposure limits. Its relevance is to delimit the present study: a deterministic hierarchy model does not represent prediction uncertainty or establish robustness merely because its computed schedules meet prescribed maintenance limits.

The recent literature also includes a multi-day maintenance-routing study (Saltzman & Stern, 2022), robust routing with heterogeneous maintenance tasks (Zhang, Chung, Ma, & Sun, 2024), and coordination between airlines and maintenance service providers (Zhang et al., 2025). These topics are important checks on the scope of any novelty claim concerning long horizons, heterogeneous obligations, or shared maintenance resources. The existence of these publications is established, but their titles alone do not determine their reset semantics or justify marking a feature as absent.

Draft placeholder: Complete primary-text review of the 2022, 2024, and 2025 studies before finalizing the feature matrix or asserting a gap relative to them. Screen the identified 2026 collaborative/predictive-maintenance article and verify availability at the review cutoff.

The proposed contribution is therefore narrower: a compact assignment-based representation of an ordered maintenance hierarchy, coupled to heterogeneous exposure measures and supported by justified formulation improvements. Its distinction from earlier models must be established at the level of reset relationships, initial-state treatment, maintenance timing, and the resulting constraints. Adding a check-type index is not sufficient. The A/B/C/D labels describe the modeled nested regime; they should not be interpreted as a universal maintenance program for all operators.

2.3 Comparable methods and evidence levels

Two evidence levels structure the evaluation. The direct computational level compares the baseline compact formulation, its optimized variant, and a semantically aligned network formulation on identical inputs. Each treatment uses the same solver configuration, time budget, and validated start policy. Reported outcomes distinguish construction effort, relaxation bounds, feasible incumbents, and optimality certification. A-only and full-hierarchy runs instead measure the cost of changing the maintenance regime; they are not equivalent formulations and cannot establish a strengthening speedup.

The literature level compares problem scope and modeling mechanisms. Published runtimes on different flight schedules, objectives, hardware, or maintenance policies provide context but not paired performance evidence. An implementation of a general network or route-selection paradigm is also not a reproduction of a particular published algorithm. Where source code, benchmark data, or full methodological details are unavailable, the comparison must state that limitation. Finally, a smaller model, a stronger relaxation, and a faster integer solve are distinct outcomes. The computational study must test each outcome rather than treating one as proof of the others.

3. Generalized maintenance-aware tail assignment

Draft placeholder: This section specifies the agreed completion-based model. It is a new connection-arc formulation, not the currently implemented daily model. Implementation, independent replay, and computational validation remain pending; no historical results are attributed to these equations.

3.1 Scope and operational conventions

Let $\mathcal{F}$ be a fixed flight timetable, $\mathcal{P}$ the individual aircraft, and $[0, T]$ the planning horizon in minutes. Flight i has fixed departure s_i, arrival e_i, duration $b_i > 0$, and endpoint airports. Aircraft j has a known initial airport, flight exposure ϕ_{jc}, and calendar age ψ_{jc}. All aircraft are available at time zero; checks already in progress are outside this first formulation. Every flight must be covered, and flight times, crew decisions, and disruptions are not optimized. The set X contains compatible flight–aircraft assignments.

Maintenance classes $\mathcal{C} = \{A, B, C, D\}$ have positive durations δ_c, eligible stations, and specified reset sets. Under the nested regime, $D \succ C \succ B \succ A$: a completed heavier intervention resets its own and all lighter obligations. This hierarchy is an input assumption about the modeled maintenance program, not universal regulatory practice. Classes A/B have flight-minute limits L_c; C/D have calendar limits $Q_c = 1440 q_c$. Initial calendar ages are converted to minutes using the same convention.

A maintenance start is a continuous decision within a ground interval, and completion is its start plus the selected duration. Counters reset only at completion. Station capacity limits simultaneous checks, including their complete occupied intervals. Selected checks must finish within the horizon. Turnaround and maintenance may overlap: a later departure must satisfy both requirements, but their durations are not simply added. An aircraft may remain grounded after a calendar deadline and complete the required check before operating again. Calendar compliance is required through flight arrival. An idle aircraft need not receive a terminal check merely because its age exceeds the threshold at the horizon end.

3.2 Finite events and aircraft paths

Create optional maintenance-event copies $\mathcal{E}_j$ for each aircraft. With $\delta_{\min} = \min_c \delta_c > 0$, the bound $|\mathcal{E}_j| = \lfloor T/\delta_{\min} \rfloor$ suffices: disjoint selected checks all finish within $[0, T]$ and each occupies at least $\delta_{\min}$ minutes. This bound avoids an arbitrary event cap, although it can be large. A smaller cap requires a validity argument or must be reported as a restriction.

Aircraft j has a graph comprising a source, compatible flight nodes, optional maintenance nodes, and a sink. Variable x_{ij} assigns a flight, v_e activates an event, and a_{juv} selects a connection from node u to node v. Exact-class decisions u_{ec}

and station decisions r_{ek} satisfy

$$\sum_c u_{ec} = v_e, \qquad \sum_{k\in\mathcal{M}} r_{ek} = v_e, \tag{1}$$

with incompatible class–station pairs excluded. Event start and completion times satisfy

$$0 \leqslant S_e \leqslant E_e \leqslant T v_e, \qquad E_e - S_e = \sum_c \delta_c u_{ec}. \tag{2}$$

Thus an inactive event has zero duration; an active event has one class, one station, and its full continuous duration.

Flight coverage requires $\sum_{j:(i,j)\in X} x_{ij} = 1$. Incoming and outgoing connection degrees both equal x_{ij} at a flight node and v_e at an event node. Source outdegree and sink indegree are one; there are no connections entering the source or leaving the sink. The direct source–sink connection accommodates an unused aircraft.

For a selected connection, the successor starts no earlier than the predecessor completes:

$$S_v \geqslant E_u - T(1 - a_{juv}). \tag{3}$$

Flight times are constants, source completion is zero, and sink start is T. Since every selected activity has positive duration, summing the selected temporal inequalities around a directed cycle is impossible. Together with degree balance, this gives a source–sink activity path, rather than disconnected cycles.

Connections preserve locations. If D_{uk} and O_{vk} are the destination and origin indicators for airport k, impose

$$a_{juv} \leqslant 1 - D_{uk} + O_{vk}, \qquad k \in \mathcal{A}. \tag{4}$$

For flights these are fixed indicators; for maintenance nodes both equal r_{ek}. Source location is the initial airport, and sink connections do not require a prescribed final airport. Source–event connections therefore allow necessary maintenance before the first flight without introducing an artificial preceding arrival.

Flights whose extended intervals $[s_i, e_i + \Delta t)$ intersect cannot be assigned to the same aircraft. Pairwise exclusions or interval-clique rows enforce the minimum turn time even when maintenance lies between two flights. Path timing also requires the next flight to wait for maintenance completion. These two conditions implement overlapping turnaround and maintenance requirements.

3.3 Completion-based exposure propagation

Let $g_{ec} = \sum_{q:\, q \text{ resets } c} u_{eq}$ indicate that event e qualifies for obligation c. This indicator describes a selected class, not repeated daily occupancy. The path provides the predecessor relation along which state changes are propagated.

For A/B, let h_{jnc} be flight exposure after activity node n, with source value ϕ_{jc}. For every selected connection into flight i,

$$h_{jic} = h_{juc} + b_i, \qquad h_{jic} \leqslant L_c. \tag{5}$$

For a connection into event e, the state is zero when $g_{ec} = 1$ and otherwise equals its predecessor's exposure. No flight exposure is accumulated during ground time. An initially overdue aircraft therefore needs a qualifying intervention before it can operate a flight, while a nonqualifying check cannot remove that obligation.

For C/D, let R_{jnc} be the most recent qualifying completion timestamp after node n. Its source value is $-1440\psi_{jc}$. Across a flight, the timestamp is unchanged and the operational age restriction is

$$e_i - R_{juc} \leqslant Q_c. \tag{6}$$

Age increases monotonically over the fixed flight interval, so compliance at arrival ensures compliance throughout that operation. After event e, set $R_{jec} = E_e$ if the selected class resets c, and otherwise retain R_{juc}. The reset occurs at completion, never at S_e.

No calendar-limit inequality is imposed on a grounded event merely because its completion is overdue. This implements the selected operational policy: overdue ground time is permitted, but the next flight must follow a qualifying completion. Likewise, no final check is forced for an aircraft that remains idle. These are boundary assumptions, not implications that follow automatically from an airline regulation.

Here is an explicit linearization. Assume nonnegative initial ages and exposures. Suppressing aircraft and obligation indices, bound each activity exposure by $0 \leqslant h_n \leqslant U$, where $U = \max\{\phi, L\}$, and each timestamp by $-\eta \leqslant R_n \leqslant T$, where $\eta = 1440\psi$ and $W = T + \eta$. For an arc into flight i, impose

$$-(U + b_i)(1 - a_{jui}) \leqslant h_i - h_u - b_i \leqslant (U + b_i)(1 - a_{jui}), \tag{7}$$

$$h_i \leqslant L, \tag{8}$$

$$-W(1 - a_{jui}) \leqslant R_i - R_u \leqslant W(1 - a_{jui}), \tag{9}$$

$$e_i - R_u \leqslant Q + \max\{0, e_i + \eta - Q\}(1 - a_{jui}). \tag{10}$$

The exposure rows apply to A/B and the timestamp rows to C/D. Flight-state bounds also apply to inactive flight copies; the relaxed transition rows leave those copies free to take bounded dummy values.

For every event, and every potential incoming arc, the A/B rows are

$$h_e \leqslant U(1 - g_{ec}), \tag{11}$$

$$-U(1 - a_{jue} + g_{ec}) \leqslant h_e - h_u \leqslant U(1 - a_{jue} + g_{ec}). \tag{12}$$

Thus a qualifying event resets its post-completion state independently of which incoming arc is selected. A nonqualifying selected event inherits exactly its selected predecessor's state. For C/D, use

$$-W(1 - g_{ec}) \leqslant R_e - E_e \leqslant W(1 - g_{ec}), \tag{13}$$

$$-W(1 - a_{jue} + g_{ec}) \leqslant R_e - R_u \leqslant W(1 - a_{jue} + g_{ec}). \tag{14}$$

When a reset is not selected, the first row relaxes over the entire timestamp range. When a reset is selected, it identifies the reset timestamp with completion. These bounds do not impose compliance while grounded. Source values remain fixed, and no sink-state compliance row is needed. All constants follow from declared variable bounds, rather than an arbitrary global numerical constant. Numerical scaling and the implementation of these rows still require validation.

This event-state formulation does not reinterpret the existing split day-pair C13 rows as completion equations. Those rows remain part of the historical daily model, with its declared interval conventions. The new model instead checks cumulative flight exposure on each operated flight along the selected path. Comparing the formulations requires explicit attention to their differing timing and feasible-region assumptions.

3.4 Continuous-time station resources and objective

Station k has B_k interchangeable resource slots. Event e selects one slot through binary $w_{ek\ell}$, where $\sum_\ell w_{ek\ell} = r_{ek}$ and $\sum_{k,\ell} w_{ek\ell} = v_e$. For two events e, f and one slot, binary $\beta_{efk\ell}$ chooses the temporal order:

$$S_f \geqslant E_e - T(3 - w_{ek\ell} - w_{fk\ell} - \beta_{efk\ell}), \tag{15}$$

$$S_e \geqslant E_f - T(2 - w_{ek\ell} - w_{fk\ell} + \beta_{efk\ell}). \tag{16}$$

If both events use the slot, one order is enforced; otherwise both rows relax. Half-open occupancy $[S_e, E_e)$ permits a slot to be reused at the exact preceding completion time.

This representation is exact for interchangeable-slot interval capacity. Nonoverlapping slot schedules imply that no more than B_k events are simultaneous. Conversely, intervals with maximum overlap at most B_k can be assigned to B_k slots by interval coloring. The interpretation does not include different manpower skills, opening calendars, or class-specific resource vectors.

The objective is

$$\min \sum_{(i,j)\in X} c_{ij} x_{ij} + \sum_{e,c} \kappa_c u_{ec}, \tag{17}$$

where κ_c is a declared per-intervention cost or penalty. Charging selection counts each check once, irrespective of the number of occupied days. Its calibration remains an experimental input; the daily implementation's duration-rounded penalty must not be presented as the same objective without explicit alignment.

3.5 Formulation size and comparison implications

The model has assignment variables, optional event selections, connections, states, and station-slot ordering decisions. If N_j is the number of activity nodes for tail j, a dense connection domain has $O(\sum_j N_j^2)$ binaries. State dimensions are $O(\sum_j N_j|\mathcal{C}|)$. With E total optional event copies, station ordering can contribute $O(E^2 \sum_k B_k)$ binaries and inequalities before reductions. This is polynomial, but may be substantially larger than the historical assignment-only model.

Optimization must therefore address both event-domain growth and connection or resource symmetry. Selecting fewer maintenance classes changes the problem, whereas removing provably impossible connections is an exact reduction. A lower runtime on a restricted event domain is not evidence of an equivalent strengthening. The computational study must separately measure the cost of continuous-time expressiveness and the benefits of valid formulation improvements under unchanged assumptions.

4. Formulation optimization

Draft placeholder: These are formulation-level validity arguments for the proposed model, not implemented optimizations or measured performance results. Their conditions must be tested before the computational comparison.

4.1 Compatible domains and impossible connections

The richer event representation requires careful control of its connection domain. A reduction is exact when it removes a decision that cannot occur in any schedule satisfying Section 3. Removing an assignment because a tail is incompatible

with a flight is one example; removing it because a particular heuristic failed to find a route is not. Likewise, excluding a check class from a station is exact only when the input maintenance program rules out that class–station combination.

Fixed-endpoint flight connections can be removed when their airports disagree or their times cannot satisfy precedence. For a direct flight connection $i \to h$, the necessary condition is $s_h \geqslant e_i + \Delta t$. For a path that inserts maintenance between these flights, the relevant conditions remain flight turnaround and completion precedence, not the sum of turnaround and maintenance durations. Consequently, deleting a maintenance opportunity because $s_h - e_i < \Delta t + \delta_c$ would be invalid under the selected overlap policy. A class can be excluded from an event bounded to that interval only when its duration exceeds the available interval itself.

For general nodes with valid time bounds, an arc $u \to v$ is impossible if $\underline{E}_u > \overline{S}_v$. The proof is immediate: every selected arc requires $S_v \geqslant E_u$, which would contradict those bounds. The bounds must hold for every admissible station and class of the node. A station-dependent bound cannot be used to remove the entire arc unless all eligible station choices fail. This distinction matters for optional events whose times and locations are not fixed in advance.

Reachability from the initial location may support further pruning, but only through a conservative graph containing every feasible connection. Absence from such a graph certifies impossibility; absence from a selected station-wait itinerary does not. All reductions must be applied consistently to degree, location, timing, and state-propagation rows. A sparse connection set with an outdated state-transfer domain is not a valid implementation of the reduced model.

4.2 Bound-derived implication constants

For a precedence row between nodes with valid unconditional bounds, use

$$M_{uv} = \max\{0, \overline{E}_u - \underline{S}_v\}, \qquad S_v \geqslant E_u - M_{uv}(1 - a_{juv}). \tag{18}$$

When the arc is selected, the required precedence is unchanged. When it is inactive, the smallest allowed successor start and largest allowed predecessor completion satisfy the relaxed row. This proves integer validity. Compared with a larger valid constant and the same variable bounds, the smaller constant also produces a nested, no-weaker LP feasible region. It does not prove a strictly better LP objective, nor a shorter branch-and-bound run on every instance.

For inactive events, time is fixed to zero by Equation (2). Thus an unconditional lower bound on their start must still include zero. A positive lower bound that holds only when an event is active cannot be substituted into Equation (18) without additional activation logic. Source, sink, and fixed flight times permit particularly transparent arc-specific constants.

State rows require different ranges. Flight exposure uses $U_{jc} = \max\{\phi_{jc}, L_c\}$, not just the regulatory threshold, because an initially overdue grounded aircraft is permitted. Calendar timestamp transfer uses $T + 1440\psi_{jc}$, while an inactive flight-age row needs only $\max\{0, e_i + 1440\psi_{jc} - Q_c\}$. These distinctions preserve the approved boundary policies. Replacing all constants by a single small value risks excluding valid initial states; replacing them by an enormous value preserves neither good numerical scaling nor a useful relaxation.

4.3 Interval-clique overlap inequalities

For a tail j, consider compatible flight intervals $I_i = [s_i, e_i + \Delta t)$. Any set $\mathcal{Q}$ with a common occupied time gives the valid inequality

$$\sum_{i \in \mathcal{Q}:(i,j) \in X} x_{ij} \leqslant 1. \tag{19}$$

The aircraft cannot operate two of these flights while satisfying minimum turnaround, regardless of any inserted maintenance. Half-open endpoints are essential: a departure exactly at the preceding arrival plus turnaround is allowed and must not be classified as an overlap.

For interval graphs, each pairwise conflict lies in a maximal clique. Therefore imposing all maximal-clique rows preserves the integer assignment restrictions of the complete pairwise system and implies its LP rows. Three mutually intersecting intervals demonstrate the potential strictness: the fractional assignment $x_{ij} = 1/2$ for all three satisfies every pairwise inequality but violates their clique inequality. This is a local relaxation relationship, not a claim that the entire model's optimal LP bound must improve. Other constraints can already exclude that point, and objective coefficients may make it irrelevant.

A sweep of interval endpoints can identify maximal cliques, processing interval endings before starts at coincident times. Each recorded clique must be restricted to assignments compatible with the particular tail. Maintenance events have variable times, so this fixed-flight construction cannot simply be reused as a complete maintenance-capacity model. Their capacity remains governed by the slot disjunctions.

4.4 Identical-event symmetry

Optional event copies for one tail have identical class/station domains and costs in the baseline design. Their labels carry no operational meaning. Number them $1, \dots, K$ and impose

$$v_{r+1} \leqslant v_r, \tag{20}$$

$$S_{r+1} \geqslant E_r - T(1 - v_{r+1}), \qquad r = 1, \dots, K-1. \tag{21}$$

Every physical schedule can be relabeled with its selected events first, in chronological order, followed by inactive copies. Events on a tail are serial, so this relabeling satisfies both rows and preserves flight assignments, event classes, locations, and cost. Conversely, the rows remove no requirement from the base model. They preserve projected operational schedules, although they intentionally remove equivalent labeled solutions.

This argument depends on interchangeability. Copy-specific time windows, class domains, or penalties can invalidate chronological relabeling. Resource-slot labels create a separate symmetry; breaking it requires its own validity argument, not an arbitrary choice of bay for each event. The computational ablation should identify which symmetry is removed and measure whether the additional rows repay their construction cost.

4.5 Exact improvements versus restrictive settings

The duration-derived event bound in Section 3 is exact under serial positive-duration checks wholly inside the horizon. An unexplained smaller event cap is restrictive. Reducing eligible check types changes the maintenance program; a short deferral window removes possible timings; requiring a specific station-wait itinerary can remove valid routes. Such options may be useful operational scenarios, but they are not equivalent strengthening techniques.

The experiment must distinguish reductions in rows or variables, changes in LP bounds, and changes in integer-solver performance. The first can be counted directly, the second requires matched relaxations, and the third requires controlled runs including timeouts and failed instances. Historical daily-model results cannot establish these effects for the completion-based formulation. Its optimized and unoptimized variants must share the same input requirements, objective, and replay validation.

5. Computational evaluation

5.1 Instances and reproducible protocol

Draft placeholder: Target for this section: 1,550 words. Report generator, multiple instance indices, maintenance pressure, solver/version, hardware, threads, budgets, source/input hashes, and warm-start policy.

5.2 Maintenance generalization and ablation

Draft placeholder: Separate changes in maintenance requirements from equivalent formulation ablations. Report construction, solver, and total times, size, memory, LP bounds, incumbents, gaps, solved counts, and timeouts.

5.3 Comparison with alternative formulations

Draft placeholder: Compare the baseline and optimized compact formulations with a semantically aligned network formulation. Include advanced published methods only at the evidence level actually available.

5.4 Scaling and sensitivity

Draft placeholder: Study fleet, horizon, maintenance duration, thresholds, and capacity. Do not combine capped and unrestricted runs or different LP policies. Populate tables and figures only from audited raw records.

6. Discussion

Draft placeholder: Target: 500 words. Answer the research questions against verified literature. Distinguish compactness, LP strength, and integer-solve performance. Discuss synthetic data, operational assumptions, comparator availability, and limited transferability. Do not claim universal regulatory coverage or solver superiority.

7. Conclusions

Draft placeholder: Target: 300 words. Summarize the verified generalization, justified optimization techniques, and matched comparative findings. Do not introduce new results or repeat the separate counterexample paper. Identify only directly relevant future work.

Authors' contribution:

Draft placeholder: Insert the actual authors' agreed contributions.

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References

Glomb, L., Liers, F., & Rösel, F. (2023). A stochastic optimization approach for optimal Tail Assignment with knowledge-based predictive maintenance. *CEAS Aeronautical Journal*, 14(3), 715–728. 10.1007/s13272-023-00663-0

Khaled, O., Minoux, M., Mousseau, V., Michel, S., & Ceugniet, X. (2018). A compact optimization model for the tail assignment problem. *European Journal of Operational Research*, 264(2), 548–557. 10.1016/j.ejor.2017.06.045

Saltzman, R. M., & Stern, H. I. (2022). The multi-day aircraft maintenance routing problem. *Journal of Air Transport Management*, 102, 102224. 10.1016/j.jairtraman.2022.102224

Torres Sanchez, D., Boyacı, B., & Zografos, K. G. (2020). An optimisation framework for airline fleet maintenance scheduling with tail assignment considerations. *Transportation Research Part B: Methodological*, 133, 142–164. 10.1016/j.trb.2019.12.008

Zhang, Q., Chung, S.-H., Ma, H.-L., & Sun, X. (2024). Robust aircraft maintenance routing with heterogeneous aircraft maintenance tasks. *Transportation Research Part C: Emerging Technologies*, 160, 104518. 10.1016/j.trc.2024.104518

Zhang, Q., Zhang, X., Ma, H.-L., Sun, X., Chan, H.-K., & Chung, S.-H. (2025). Synergy of airline and maintenance service provider in aircraft maintenance routing. *Transport Policy*, 166, 32–49. 10.1016/j.tranpol.2025.02.022

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ІЄРАРХІЧНЕ ТЕХНІЧНЕ ОБСЛУГОВУВАННЯ В ПРИЗНАЧЕННІ ЛІТАКІВ: ФОРМУЛЮВАННЯ ТА ОПТИМІЗАЦІЯ

Draft placeholder: Prepare a Ukrainian abstract consistent with the completed English abstract, meeting the journal's length requirements.

Ключові слова: ***цілочислове програмування, маршрутизація літаків, накопичене навантаження, календарні обмеження, допустимі нерівності, обчислювальний аналіз*** Автори заявляють про відсутність конфлікту інтересів. Спонсори не брали участі в розробленні дослідження (у зборі, аналізі чи інтерпретації даних, якщо це мало місце), у написанні рукопису та в рішенні про публікацію результатів.

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